

Databricks Lab : Single Node Cluster Setup

Duration : 60 Min.

Scenario

As a Junior Cloud Engineer at LabsKraft, you are tasked with provisioning a development compute environment in Databricks. Your objective is to deploy a Databricks cluster configured as a single-node system with Photon acceleration disabled to save development costs.

Input Details

The environment has been pre-configured with the following resources:

- **Workspace:** Pre-configured through KodeBuck Service credentials (`DATABRICKS_HOST`, `DATABRICKS_TOKEN`)
- **Resource Group:** `rg-iRUN-LTM-Assessment`

Username & Naming Conventions

Your candidate prefix is a combination of your username (the part of your email/login before the @ or _ symbol) and exam code. For example, if your username is student (or `student_labskraft.com`) and your exam code is exam123, your prefix is **student-exam123**.

You must name your resources accordingly:

- **Cluster Name:** `<prefix>-db-cluster` (e.g. `student-exam123-db-cluster`)

Task Objectives

Perform the following configuration steps directly in the Databricks Workspace Console or using the Databricks API:

1. Create Databricks Cluster

- **Cluster Name:** `<prefix>-db-cluster` (e.g. `student-exam123-db-cluster`)
- **Databricks Runtime Version:** Set the runtime to 17.3 LTS (`17.3.x-scala2.13`).
- **Node Type:** Set the node type to `Standard_F4`.
- **Single Node Cluster:** Enable Single Node mode (which sets worker count to 0 and uses the `singleNode` cluster profile).
- **Photon Acceleration:** Ensure Photon Acceleration is **Disabled**.
- **Autotermination:** Set the cluster to auto-terminate after 30 minutes of inactivity.

Verification

Once you have performed the tasks, you can run the verification script to check your progress and receive your score. The verification system will check the actual resources in the Databricks Workspace.

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	Cluster existence (<prefix>-db-cluster exists)	4 Marks
TC2	Spark Runtime Version (17.3.x-scala2.13)	4 Marks
TC3	Autotermination Verification (30 minutes)	4 Marks
TC4	Single Node Mode Enabled (0 workers, singleNode profile)	4 Marks
TC5	Node Type (Standard_F4) & Photon Disabled	4 Marks

Total Score: 20 Marks

Important Notes

- Ensure all resource names contain your candidate username prefix in lowercase.
- The cluster name must follow Databricks naming rules.